

HARINE S/O SUMEN

harinesumen@gmail.com | +65 90294767 | [linkedin.com/in/harinesumen](https://www.linkedin.com/in/harinesumen) | github.com/unplayable01

PROFESSIONAL SUMMARY

Computer Science undergraduate at SIT / DigiPen with an Electronic and Computer Engineering diploma, driven by how software works under the hood. Tech lead on a custom C++ engine for a published game; owned the Go backend and Jenkins CI/CD for an Android vision app. Also a teacher, chess player and composer, seeking software engineering roles.

EDUCATION

Singapore Institute of Technology, in collaboration with DigiPen Institute of Technology Singapore *Sep 2024 – Present*
BSc (Hons) in Computer Science in Real-Time Interactive Simulation *Expected Graduation: Aug 2028*

Ngee Ann Polytechnic *Apr 2019 – May 2022*
Diploma in Electronic and Computer Engineering

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Go, Kotlin, Java, TypeScript, Lua, GDScript, VBA

Engine & Game Development: ECS Architecture, OpenGL, ImGui, Lua Scripting, Memory Management, Input Handling

Backend & Infrastructure: Go + SQLite APIs, Jenkins CI/CD, Linux Servers, Nginx, Git LFS, Gitea, Tailscale

Tools: GitHub, Visual Studio, CMake, Godot, Android Studio, Jira, FMOD, Aseprite, DaVinci Resolve, Microsoft Access VBA

Additional: Software Architecture, Test Automation, Documentation, Calculus, Leadership, Teaching, Music Composition

PROJECTS

Technical Lead | **TTBEngine** — Custom C++ Engine for TiltTiltBoom (Published Game) *Sep 2025 – May 2026*

- With 8 programmers, designers and artists sharing one codebase, designed **TTBEngine's** modular C++ ECS architecture with an ImGui editor and hot-reloaded Lua scripting, so the engine adapted as scope grew; top committer.
- To make **TiltTiltBoom's** screen-tilting bomb-defusal mechanic feel precise, built the controller system (gamepads, two-axis gyro aiming, haptics) and a deterministic replay system that powers per-level leaderboards.
- Built CMake targets and an Inno Setup installer to ship the game publicly; also composed its original soundtrack.

Backend & Infrastructure Owner | **PinPoint** — Android Bowling Coach App *2026*

- The app needed secure accounts and private video storage, so built a **Go + SQLite API server** with invite-only crypt-hashed accounts, token sessions, upload limits and rate limiting, covered by 11 test files.
- To keep broken builds from reaching users, set up **Jenkins CI/CD** that builds signed APKs and deploys behind go vet/go test gates, rolling back automatically if the server is unhealthy; hardened the VM with HTTPS, Tailscale and fail2ban.

Creator | **Aaron** — Python Webcam Security Guard (Open Source) *Sep 2026*

- To keep an unattended PC safe, built **Aaron**, a webcam guard that recognizes its owner's face (OpenCV) and locks Windows for strangers, controlled via Telegram, offline voice wake (Vosk) and MediaPipe gestures.
- Telegram commands raced the camera loop, so split Aaron into three processes via file-based IPC, ending the race.

WORK EXPERIENCE

Teaching Assistant | DigiPen Institute of Technology Singapore (Hybrid) *Jan 2026 – Apr 2026*

- Juniors new to university maths struggled with abstract calculus, so ran crash courses for 50+ students per session and 1-on-1 consultations, breaking concepts into step-by-step explanations until each one clicked.

Software Engineer Intern | Denso Wave Incorporated *Oct 2021 – Feb 2022*

- Manual inventory tracking and microcontroller testing slowed the team, so built VBA inventory management systems and Python automation scripts, reducing manual testing.
- Debugged, tested and documented processes with cross-functional engineers, improving internal tools.

LEADERSHIP & ACTIVITIES

President & Quartermaster | SIT Mindsports *Jan 2025 – Dec 2025*

- Balancing club leadership, equipment and logistics with competing in 10 chess competitions, helped the team place **3rd Runner-Up** in Singapore's Intervarsity League against all universities, polytechnics and JCs.

FYE Helix Assistant Head Facilitator | Singapore Institute of Technology *May 2025 – Aug 2026*

- Freshmen needed help settling into university life, so co-led the facilitator team running FYE orientation activities.